Behavioral Science and Climate Policy

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Summary and Keywords

Behavioral science consists of the systematic analysis of processes underlying human behavior through experimentation and observation, drawing on knowledge, research, and methods from a variety of fields such as economics, psychology, and sociology. Because policymaking involves efforts to modify or alter the behavior of policy-takers and centers on the processes of decision-making in government, it has always been concerned with behavioral psychology. Classic studies of decision-making in the field derived their frameworks and concepts from psychology, and the founder of policy sciences, Harold Lasswell, was himself trained as a behavioral political scientist. Hence, it should not be surprising that the use of behavioral science is a feature of many policy areas, including climate change policy.

This is given extra emphasis, however, because climate change policymaking and the rise of climate change as a policy issue coincides with a resurgence in behaviorally inspired policy analysis and design brought about by the development of behavioral economics. Thus efforts to deal with climate change have come into being at a time when behavioral governance has been gaining traction worldwide under the influence of works by, among others, Kahneman and Tversky, Thaler, and Sunstein. Such behavioral governance studies have focused on the psychological and cognitive behavioral processes in individuals and collectives, in order to inform, design, and implement different modes of governing. They have been promoted by policy scholars, including many economists working in the area who prefer its insights to those put forward by classical or neoclassical economics.

In the context of climate change policy, behavioral science plays two key roles—through its use of behaviorally premised policy instruments as new modes of public policy being used or proposed to be used, in conjunction with traditional climate change policy tools; and as a way of understanding some of the barriers to compliance and policy design encountered by governments in combating the “super wicked problem” of climate change. Five kinds of behavioral tools have been found to be most commonly used in relation to climate change policy: provision of information, use of social norms, goal setting, default rules, and framing. A large proportion of behavioral tools has been used in the energy sector, because of its importance in the context of climate change action and the fact that energy consumption is easy to monitor, thereby facilitating impact assessment.

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Introduction

Climate change is defined as an identifiable change in the state of the climate (e.g., by using statistical tests) that persists for an extended period of time, typically a decade or longer. This change in the state of the climate may be evidenced through changes in mean temperatures or changes in the variability of its properties or both (Intergovernmental Panel on Climate Change [IPCC], 2014A). In the face of scientific agreement regarding the role of greenhouse gases in climate change, the focus of the United Nations Framework Convention on Climate Change (UNFCCC) has been on stabilizing greenhouse gas concentrations to levels that "would prevent dangerous anthropogenic interference with the climate system" (UNFCCC, 1992, p. 201).

Climate change policy tended at first to focus on mitigation efforts, with adaptation policy being under-emphasized in many countries (Stern, 2007), based on the assumption that significant adaptation measures could be avoided if mitigation was effective (Schroeder, 2010).[1] But as the dominant view changed to regard some degree of climate change as inevitable, notwithstanding the success of mitigation measures, countries started to focus on adaptation alongside mitigation (Keskitalo, 2010). This was most apparent at the 2015 Paris Conference of the Parties (COP), where 146 out of 162 actors included adaptation components within their climate target pledges, referred to as the "Intended Nationally Determined Contributions" (Tobin, Schmidt, Tosun, & Burns, 2018).

When it comes to mitigation policy, a majority of mitigation actions have centered on lowering of emissions through energy saving and replacing fossil fuels with renewable energy (Fleig, Schmidt, & Tosun, 2017). Examples of the former have included promoting more energy efficient applications (Lipscy & Schipper, 2013) and an example of the latter is through promoting new technologies (Lechtenböhmer & Luhmann, 2013; Pacala & Socolow, 2004).

Adaptation policies on the other hand, have tended to focus on activities such as developing strategies for water management (using scarce water resources efficiently, managing droughts and flooding), adapting structures such as buildings to global temperature rises, and dealing with extreme weather events (Fleig et al., 2017).

While mitigation benefits may be more global in scale and lag in time, adaptation benefits tend to be more localized and are realized more quickly (Wilbanks et al., 2003). The multifaceted character of mitigation and adaptation (Buuren et al., 2013) feeds into the complexity, diversity, externality, and uncertainty associated with climate change, making it one of the major "wicked" problems in contemporary times (Head & Alford, 2015; Levin, Cashore, Bernstein, & Auld, 2012). Four key features that characterize climate change, combine to create a policymaking "tragedy,"—a situation where traditional techniques may be inadequate in finding solutions. These are that time is running out; that those seeking to find a solution to the problem are also those causing it; that there is no central

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authority to address it; and that, partly as a result, policy responses discount the future irrationally (Head & Alford, 2015; Levin et al., 2012).

Responding to climate change, according to the *World Development Report 2010: Development and Climate Change* (World Bank, 2010), is difficult because of three sources of inertia. The first is inertia of the environment and climate system (because of the lag in emissions reductions and corresponding changes in concentrations and temperature). The second is inertia embodied in physical capital (due to difficulties in shifting from a high-carbon to a low-carbon capital stock, and the lag in research and development efforts and their subsequent deployment). The third is inertia in the behavior of organizations and individuals (i.e., the patterns in organizational and individual behavior that affect implementation of policies and hamper their success). It is in the context of the latter, particularly individual behavior, that behavioral science provides a way to understand barriers to compliance and the behavioral processes that are operating through the climate policy cycle.

Thus, behavioral science is important because, first, climate change is affected by myriad acts of private consumption especially in case of emissions in developed countries. Second, larger processes of change in organizations, institutions, and political systems are driven by individuals. And third, decision-makers at the different stages of the policy cycle may be applying the same mental processes as other individuals (World Bank, 2010).

Behavioral science consists of the systematic analysis of processes underlying human behavior through experimentation and observation, drawing on knowledge, research, and methods from a variety of fields such as economics, neuroscience, psychology, and sociology, among others (Sousa Lourenço, Ciriolo, Rafael Almeida, & Troussard, 2016). Since policymaking involves efforts to modify or alter the behavior of policy-takers and centers on the processes of decision-making in government, it has always been concerned with behavioral psychology. Classic studies of decision-making in the field (Lindblom, 1959) derived their frameworks and concepts from psychology; the founder of the policy sciences, Harold Lasswell, was himself trained as a behavioral political scientist (Torgerson, 1985).

Hence, it should not be surprising that the use of behavioral science is a feature of many policy areas, including climate change policy. This is given extra emphasis, however, because climate change policymaking and the rise of climate change as a policy issue coincides with a resurgence in behaviorally inspired policy analysis and design brought about by the development of behavioral economics (see Bavel, Herrmann, Esposito, Proestakis, & Institute for Prospective Technological Studies, 2013; Behavioural Insights Team [BIT], 2010, 2014; Centre d’Analyse Stratégique, 2011; Datta & Mullainathan, 2014; Halpern, 2015; Lunn, 2013, 2014; Sunstein, 2013, 2016).

That is, efforts to deal with climate change have come into being at a time when behavioral governance has been gaining traction worldwide under the influence of works by, among others, Kahneman and Tversky (Kahneman & Tversky, 1979, 2000; Tversky & Kahneman, 1973, 1974, 1981); Camerer (Camerer, 1999, 2004; Camerer, Issacharoff, Loewenstein, O’Donoghue, & Rabin, 2003); Banerjee and Duflo (Banerjee & Duflo, 2011; Duflo,

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Kremer, & Robinson, 2011); and Thaler and Sunstein (Sunstein, 2013, 2014, 2015, 2016; Sunstein & Thaler, 2003; Thaler & Sunstein, 2008).

Behavioral governance studies have focused on the psychological and cognitive behavioral processes in individuals and collectives in order to inform, design, and implement different modes of governing (Strassheim & Korinek, 2015) and have been promoted by many policy scholars, including many economists working in the area who prefer its insights into the actual behavior of targets to those put forward by classical or neoclassical economics (see Fehr & Schmidt, 2003; Loewenstein, 2008; Mullainathan & Shafir, 2014; Thaler, 2015; Strassheim & Korinek, 2015). Studies undertaken in this vein include the behavioral analysis of policy instruments as well as behaviorally informed studies of policy, regulation, and law (Sunstein, 2016).

In the context of climate change policy, behavioral science plays two key roles. First, behaviorally premised policy instruments have emerged as new modes of public policy being used or proposed for use in conjunction with traditional climate change policy tools. These new policy tools are based on behavioral insights related to psychological and cognitive processes in individuals that are intended to influence individuals' choices in a particular way. Such policy tools have been referred to as "nudges" based on the work inspired by Thaler and Sunstein (see for example Sunstein, 2013, 2014, 2015; Sunstein & Thaler, 2003; Thaler & Sunstein, 2008)[2] Five kinds of nudges have been found to be most commonly used in the context of climate change policy: provision of information, use of social norms, goal setting, default rules, and framing.

Second, behavioral science provides a way of understanding some of the barriers to compliance encountered by governments in combating the "super wicked problem" of climate change. Because, eventually, it is individuals who play a role through each of the stages of the climate change policy cycle, thus, by shedding light on aspects of behavioral processes, behavioral science enables the design and implementation of effective climate change policy.

This discussion proceeds in three parts. The first section ("CLIMATE POLICY'S BEHAVIORAL TOOLKIT: MOVING BEYOND TRADITIONAL POLICY INSTRUMENTS") examines the genesis of behavioral sciences in climate change policy through the use of behavioral policy tools and evaluates how they differ from traditional policy instruments. The use of behavioral instruments in climate policy comes against the backdrop of their rising influence in the policy world at large, and thus the spread of behavioral policy is briefly discussed, followed by an examination of the specific forms behavioral policy instruments have taken in the context of climate change.

The second section ("THE BEHAVIORAL BARRIERS FACED BY CLIMATE POLICY") focuses on the role of behavioral sciences in understanding barriers to compliance with climate change policy and engaging in pro-environment behavior. It also explores the behavioral processes of policymakers.

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In the third section (“CHALLENGES TO THE USE OF BEHAVIORAL SCIENCES IN CLIMATE POLICY”) the challenges that are faced in the use of behavioral techniques in climate policy are summarized and the way forward for behavioral sciences in climate policy is assessed.

Climate Policy’s Behavioral Toolkit: Moving Beyond Traditional Policy Instruments

There are many ways to categorize policy tools, but policy instruments are generally categorized into four categories—inducements (such as incentives and sanctions), regulations or tools that use coercion, tools that rely on moral suasion and information, and those that utilize government personnel to directly implement policy goals (Hood, 1986). Traditional climate change policy instruments fall into all these categories. A typology of climate policy tools could also be made based on whether they are primarily targeted toward mitigation or adaptation or both (Klein, Schipper, & Dessai, 2005).

Early discussion of climate change policy tools was centered on mitigation through instruments such as carbon taxes. Beginning in 1997 the focus was on tools to promote quantitative restrictions in emissions such as cap and trade systems (Nordhaus, 2015). These along with hybrid systems (for example, those that combine a cap and trade system with a price ceiling or floor or both) are premised on the principle of raising carbon prices through market mechanisms, so as to create economic incentives to decrease carbon emissions while allowing emitters to decide exactly how much they will reduce and by what method. One of the main theoretical benefits of economic instruments lies in their potential to reduce emissions at a lower cost as compared to other instruments such as direct regulations, which are cruder and do not allow as much discretion to emitters (Goulder & Schein, 2013; Harrison, Foss, Klevnas, & Radov, 2011).

The other modes of climate change policy include regulatory instruments, voluntary agreements, and tools to stimulate technological change. While such tools can be targeted toward both mitigation and adaption, their dominant use has been in the sphere of mitigation. Regulatory instruments include the use of non-tradable permits, technology and performance standards, and product bans (Environmental Protection Agency [EPA], 2001). Voluntary agreements can take many forms, but in general tend to be agreements between governmental agencies and private parties to achieve environmental objectives, especially carbon emission reductions (IPCC, 2014B).

Tools to stimulate technology change in the direction of low-carbon (“clean” or “green”) technologies include the use of demand-pull, supply-push, and supply-demand interface types of instruments (Leggett, 2011). Examples of demand-pull based instruments are “technology forcing” regulations, where a regulator mandates standards that cannot be met with existing technologies, such as improved automobile fuel standards (Gerard &

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Lave, 2007), and fiscal incentives that reduce the price of certain technologies for consumers, such as solar panels.

Supply-push tools include subsidies to develop new technologies such as improved electric batteries (Leggett, 2011), while supply-demand interface instruments focus on improving the interface between technology suppliers and users such as the creation of university-industry "discovery" parks (Taylor, 2008). Virtually all countries rely on the aforementioned policy tools (regulations, technology, voluntary approaches) in addition to policy instruments that are aimed at raising the prices of carbon emissions (Nordhaus, 2015).

The introduction of behavioral policy instruments into this traditional toolkit of climate change policy can be traced to the *Third Assessment Report* of the IPCC, which recognized the role of barriers to effective climate change policies, as imposed by social norms, individual habits, attitudes, and values. The report proposed the adoption of changes in behavior, lifestyle, social structure, and institutions as one among numerous policy strategies (IPCC, 2001).

A clear articulation of the critical role of behavior in climate change policy, however, came with the U.K. Stern review (2007), one of the seminal documents to estimate the costs of climate change.[3] The Stern review was among the first such studies to unequivocally emphasize the need for the removal of barriers to behavioral change as a key element essential for effective climate change policymaking (Stern, 2007).[4]

Behavioral Instruments Versus Traditional Policy Tools

Barriers to behavioral change in the context of climate change policy stem from multiple sources. They may be a result of information or psychological and cognitive influences (Zaval & Cornwell, 2016) that shape the perception of the problem and policy solutions by the policymaker as well as policy-taker. They may also be attributable to monitoring and enforcement issues; incentive issues; resource and autonomy barriers[5] (Weaver, 2009; Weaver, 2014, 2015); capacity issues (Smit & Wandel, 2006); lack of political awareness and urgency to adapt (Massey, Biesbroek, Huitema, & Jordan, 2014); values, beliefs, and norms (Matti, 2015); as well as emotions (Smith & Leiserowitz, 2014).

Behavioral policy instruments or nudges, are new modes of climate change policy that aim to reduce some of these barriers to behavioral change by incorporating behavioral insights related to human psychological and cognitive processes into their design. Their underlying rationale differs from that of traditional policy instruments.

Traditional policy tools such as inducements, regulation, and knowledge (Schneider & Ingram, 1997) are based on assumptions of the theory of rational choice and utility maximization (Stewart, 1993; Stone, 2001). Rational choice theory views individuals as perfectly rational, utility-maximizing decision-makers. Thus, the theory assumes that human agents (i) are self-interested utility maximizers; (ii) have stable and ordered preferences; (iii) have correct information to assess outcomes relative to objectives; and (iv) make

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strategic decisions based on their preferences, and calculations of risks, costs, and rewards (Jones, Boushey, & Workman, 2006).

Behavioral policy instruments on the other hand, do not take for granted the perfect rationality that is assumed by the classical model. Thaler and Mullainathan (2008) put forward three categories for agents' deviations from the classical model: (i) bounded rationality and limited cognitive abilities, (ii) limited self-interest and reciprocity, and (iii) bounded willpower. Over time such behavioral departures from the predicted utilitarian model have been documented so as to identify a number of behavioral patterns that influence individual decision-making.

One of these is the use of "heuristics," or mental shortcuts that reduce the cognitive burden associated with decision making (Shah & Oppenheimer, 2008). In their pioneering work Tversky and Kahneman identified three central heuristic principles affecting behavior—availability, representativeness, and anchoring (Tversky & Kahneman, 1974). In later work they demonstrated the influence of framing (of acts, contingencies, and outcomes) on preferences and the characteristic nonlinearities of values and decision weights (Kahneman & Tversky, 1979, 2000; Tversky & Kahneman, 1981).

Following Kahneman and Tversky, a number of other behavioral patterns that influence decision-making have been noted, such as "overconfidence" (Moore & Healy, 2008), "present bias" (O'Donoghue & Rabin, 1999), and the tendency to gravitate toward the default option (Lunn, 2014).[6] Such patterns of decision-making have been explained by drawing on concepts from psychology.

Psychology literature speaks of two systems of thinking that human agents employ—system I and system II. System I is intuitive and automatic (adopts a narrower frame and employs little or no effort), whereas system II is reflective and deliberate (adopts a wider frame and employs more effort) (World Bank, 2015). The dominance of the automatic system leads to the adoption of heuristics in making decisions (Kahneman, 2003).

Thus, an increasing awareness and recognition of these behavioral processes has contributed to the use of behavioral policy tools that aim to reduce "behavioral market failures" (Thaler & Sunstein, 2008). Such behaviorally premised policy instruments make use of insights drawn from behavioral sciences so as to secure better compliance with government aims and policies.

In the context of climate change policy, the use of behaviorally premised policy tools comes against the backdrop of their rising influence and use in the policy world in general (see Bavel et al., 2013; BIT, 2010, 2014; Centre d'Analyse Stratégique, 2011; Datta & Mullainathan, 2014; Halpern, 2015; Lunn, 2013, 2014; Sunstein, 2013, 2016).

Spread of Behavioral Policy

While the United States and United Kingdom are regarded as among the earliest and most enthusiastic adopters of behaviorally premised policy instruments (Lunn, 2014), pol-

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icymakers in the European Union (Bavel et al., 2013; Sousa Lourenço et al., 2016) and other countries like Singapore (Low, 2012), and Australia (Behavioural Economics Team of the Australian Government [BETA], 2016) have also focused on the application and implication of behavioral sciences for policymaking. In these countries, "nudge units" have been created either within the civil services, or working closely with them, for instance the Behavioural Insights Team in the UK Cabinet Office (Dolan & Galizzi, 2014).[7]

Similar in-house behavioral insights teams have come into being either at the centralized (regional) or ministerial level in the European Commission, France, Germany, Israel, the Netherlands, and South Africa (Organisation for Economic Co-operation and Development [OECD], 2017B). In other countries such as Norway and Denmark, the push for behaviorally informed interventions has come with the involvement of actors from outside of the government. For instance, not-for-profits in Norway (such as "GreeNudge") and Denmark (such as "iNudgeYou") are dedicated to improving decision-making through information dissemination, small-scale experimentation, and training (Kallbekken, Sælen, & Hermansen, 2013).

Closely tied to the spread of behaviorally premised policy has been the use of randomized controlled trials (RCTs) (Galizzi, 2017). This has extended to developing countries too, where applications of behavioral science have made use of RCTs that employ behavioral interventions to tackle poverty (see for instance Banerjee & Duflo, 2011; Karlan & Appel, 2011). The collation of information on these RCTs by evaluation databases such as the Network of Networks on Impact Evaluation (NONIE), the Coalition for Evidence-Based Policy, the World Banks' Development Impact Evaluation (DIME), the Abdul Latif Jameel Poverty Action Lab (J-PAL), and the American Economic Association (Glennerster & Takavarasha, 2013) has further contributed toward the wider dissemination and application of behavioral sciences in the policy world, including the development sector.

The mainstream recognition and acceptance of what may be termed "applied behavioral science" (Kahneman, 2013) in policymaking by international organizations is evident in documents such as the World Bank's *World Development Report 2015: Mind Society and Behavior*. The report focuses on the idea that "how humans think (the processes of mind) and how history and context shape thinking (the influence of society) can improve the design and implementation of development policies and interventions that target human choice and action (behavior)" (p. 3). Subsequently, the OECD (2017A) report titled *Behavioral Insights and Public Policy* provides the first ever comprehensive international overview of the application of more than 100 behavioral insights across the policy sector. A second OECD (2017B) publication has detailed these developments specifically in the context of tackling environmental problems. One of the key findings of the OECD reports is that the use of behavioral insights has moved beyond a trend and enjoys considerable support from leaders of public organizations.

According to Oliver (2013) the support for behaviorally informed policy comes from the search for alternatives to interventionist measures such as bans and regulations by liberal-minded politicians who are seeking ways to motivate people to change self-and-society-

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harming behavior. Thaler and Sunstein (2008) on the other hand, focus on the ability of behavioral interventions to achieve higher target compliance through relatively lower costs. This is done by changing how the problem is diagnosed, how solutions to it are designed, and how the scope of the problem is defined.

Behavioral Tools in Climate Change Policy

Behavioral insights are increasingly being applied across the world, but their application tends to be restricted to sectors in which they were first introduced (OECD, 2017A). In the case of climate change policy, most nudges have tended to focus on energy usage and efficiency because of the importance of energy policy in the context of climate change action and the fact that energy consumption is easy to monitor, thereby facilitating impact assessment through use of RCTs (OECD, 2017B). In addition to the energy sector, nudges have also been used in the water sector and for the environment at large, with some measures reflecting positive implications for adaptation as well. For instance, the use of social norm comparisons was found to reduce residential water consumption in Cape Town, South Africa, during times of water scarcity (Brick, De, & Visser, 2017; Smith & Visser, 2014).

It has been argued that nudges have the potential to meet both mitigation and adaptation goals, if they can be tailored toward policy targets after taking note of differences in attitudes, motivations, and circumstances. So, for instance, nudges at key points could provide a cost-effective policy instrument to encourage woodland creation (Valatin, Moseley, & Dandy, 2016). Woodland creation (through reforestation with native and diverse tree species) can enhance adaptation and mitigation goals simultaneously (Moser, 2012).

Of the larger menu of behavioral policy tools five kinds of nudges in particular have been commonly used as climate change policy tools. These are: (i) provision of information, (ii) use of social norms, (iii) goal setting, (iv) default rules, and (v) framing (see BIT, 2011; Nielsen et al., 2017; OECD, 2017A, 2017B; World Bank, 2015).[8]

Information

The provision of information, in the form of disclosures that are comprehensible, simple, and accessible, is used to shift individual behavior in a more pro-environment direction. This may take the form of providing individuals with information regarding the consequence or costs of certain actions or it may take the form of informing them about their own past choices and its consequences (“smart disclosure”). The latter is information that public and private institutions are often privy to but that individuals themselves lack (Sunstein, 2014).

In this regard, experimental trials by the British Office of Gas and Electricity Markets (OFGEM) that have evaluated behavioral interventions individually and in combination have found that the deployment of smart meters that provided real-time as well as historic-consumption information contributed the most to energy savings (OFGEM, 2011). Similarly, studies show that energy related labeling in the United Kingdom (Department

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of Energy and Climate Change [DECC] & BIT, 2014) and Switzerland (Stadelmann & Schubert, 2018) resulted in the purchase of more energy efficient products.

Taniguchi and Fujii (2007) found that the frequency of using public transport increased in the city of Obihiro, Japan, when information about the bus was communicated to participants in the form of a one-shot travel feedback program, although this study was based on self-reported measures. Several studies examining the impact of provision of personalized information regarding travel behavior, such as through travel carbon calculators, have found that these lead to reduced car travel or increased use of public transport (Fujii & Taniguchi, 2005; Meloni, Spissu, & Bhat, 2011).

Similar adoption effects toward more sustainable alternatives have been found for provision of information related to fuel efficiency, emissions, and energy costs (see for instance Codagnone, Bogliacino, & Veltri, 2013; Delmas, Fischlein, & Asensio, 2013; Kallbekken et al., 2013).

In countries like Estonia (Rivas, Cuniberti, & Bertoldi, 2016), South Africa, and the United Kingdom (OECD, 2017A) targeted behavioral levers have begun to be employed through provision of simplified information to consumers on their energy consumption.

Social Norms

Social psychologists regard social norms as an important dimension affecting human behavior (McKirnan, 1980; Staub, 1972). Social norms are defined as the broadly shared beliefs about what group members are likely to do and ought to do. They are an informal governance mechanism that can exert a powerful influence on decision-making (Elster, 1989). According to behavioral economists, social norms affect choices: (i) through their intrinsic value, (ii) through reputational value, and (iii) by their effect on self-image (Sunstein, 1996).

Social norms, by informing individuals about the behavior of others (and whether the behavior of others is undesirable) and then highlighting what most people think others should do, act as a nudge toward changing behavior (Sunstein, 2014). For instance, an energy program in the United States that involved providing utility consumers with information about how their energy usage compared to that of their neighbors resulted in a subsequent reduction in energy consumption (Allcott, 2011).

In the context of climate change, social norms have been successfully invoked in the energy sector (see Ayres, Raseman, & Shih, 2013; Dolan & Metcalfe, 2013) and water sector (Datta et al., 2015; Ferraro, Miranda, & Price, 2011; Ferraro & Price, 2013) and in enabling other pro-environmental behavioral change (Goldstein, Cialdini, & Griskevicius, 2008; Kuhfuss, Préget, Thoyer, & Hanley, 2016) across countries. Social norms also operate on policymakers, wherein the actions of neighboring jurisdictions have been seen to influence policy choices, for example in the case of carbon taxes (Krause, 2011).

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Although the use of social norms in behavioral policy tools has been aimed at individuals, social norms may also be applied to organizations to overcome behavioral barriers to investments in energy-saving technologies (Centre for Sustainable Energy and the Environmental Change Institute, University of Oxford, 2012), given the strong effects exerted by normative and mimetic behavior (Perez-Batres, Miller, & Pisani, 2011). While they haven't been used in an international context yet, evidence of their potential for use can be found in policy diffusion literature. For instance, Fankhauser, Gennaioli, and Collins (2016) find that the propensity to legislate on climate change increases with the number of similar laws passed elsewhere.

Goal Setting

Goal setting or getting people to pre-commit to strategies at a certain time, is used as a policy tool to better motivate action, counteract lack of willpower, and overcome the tendency to procrastinate (Sunstein, 2014). It is argued that goals also indirectly affect action by leading people to discover, desire, or use knowledge and strategies related to the task (Locke & Latham, 2002).

The greater the cost of breaking the commitment can be made, the more effective it is in achieving behavior change (Dolan, Hallsworth, Halpern, King, & Vlaev, 2010). Another related mechanism to align future behavior in the desired direction is to make the commitment public. Individuals' desire to maintain a consistent and positive self-image (Cialdini, 2008) makes them likely to uphold commitments in order to avoid reputational damage or cognitive dissonance (Festinger, 1962).[9]

Evidence of this has been found in studies where homeowners who made public commitments toward energy conservation were found to have lower rates of increase in energy use compared to a control group (Pallak & Cummings, 1976). Similarly, it was observed that residents who made written commitments to voluntary recycling were more likely to participate in recycling programs as compared to other groups (Werner et al., 1995). These studies predate the rise of behaviorally premised instruments in the climate policy toolkit, but the explicit incorporation of goal-setting as a nudge toward pro-environmental behavior in later studies has shown similar positive results.

Loock, Staake, and Thiesse (2013) find that goal-setting with a web-based energy feedback platform stimulates energy conservation. Their findings are similar to those of Abrahamse, Steg, Vlek, and Rothengatter (2007), who find that a combination of tailored information, goal-setting, and feedback resulted in significantly higher energy savings for households subjected to this treatment.

Outside of the energy sector, a field experiment in a hotel in California that examined the effect of guests' commitment to practice environmentally friendly behavior during their stay (measured through their towel usage) found it to have a significant impact on actual behavior (Baca-Motes, Brown, Gneezy, Keenan, & Nelson, 2013). In Costa Rica a RCT testing behavioral treatments for water conservation found that for low water consump-

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tion households, plan-making was the most effective in reducing consumption (Datta et al., 2015).

What is particularly promising about goal-setting as a climate policy tool is its ability to close the gap between attitude and action. In addition, goal-setting has been found to change behavior both in the short and the long term (Lokhorst, Werner, Staats, van Dijk, & Gale, 2013).

While the goal-setting nudge has been used to usher in more environment friendly behavior by individuals, at the macro level goal-setting reflects a novel approach to global governance for environmental and climate change policy. This is seen in the United Nations' sustainable development goals (Biermann, Kanie, & Kim, 2017) as well as in the Intended Nationally Determined Contributions of UNFCCC member states regarding their efforts toward climate change (Fleig et al., 2017).

Defaults

Defaults are settings that stick or outcomes that apply when individuals do not take active steps to change them (Brown & Krishna, 2004; Johnson & Goldstein, 2013). Because institutions normally define defaults, designing them in certain ways can have a profound impact on the outcomes of individual choice (Barr, Mullainathan, & Shafir, 2009).

According to Sunstein and Reisch (2013) three principal factors contribute to the large effect of defaults on outcomes: (i) suggestion and endorsement, that is, people, especially if they consider themselves non specialists, think that the default was chosen with good reason and by someone sensible (see McKenzie, Liersch, & Finkelstein, 2006); (ii) the tendency to procrastinate and the power of inertia, which lead to people continuing with the status quo (Sethi-Iyengar, Huberman, & Jiang, 2004); and (iii) defaults provide a reference point (relative to which changes are evaluated).

Examples of "climate-friendly defaults" (Sunstein & Reisch, 2016) include their use in Southern Germany where they were found to result in a significantly higher percentage of customers buying green electricity, or energy that came from an environmentally friendly source (Pichert & Katsikopoulos, 2008). Other studies have reported comparable results, such as a higher incidence of green energy use, when green energy was the default option (Ebeling & Lotz, 2015).

Much in the same manner, an RCT in Spain found higher contributions to carbon offsetting programs when the default option was the opt-out treatment (Araña & León, 2013). Yet another study exploring consumer choice between an energy efficient but costly Compact Fluorescent Light Bulb (CFLB) and an inefficient but inexpensive Incandescent Light Bulb (ILB) found a lower preference for the ICB when the energy efficient CFLB was the default (Dinner, Johnson, Goldstein, & Liu, 2011). Several municipal electricity utilities in Switzerland have changed the default electricity mix to a greener tariff (Sousa Lourenço et al., 2016)

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Framing

Framing is a cognitive bias in which individuals tend to make decisions influenced by how information is presented or framed (Tversky & Kahneman, 1981). Presenting the same information in different formats can affect people's decisions (Zaval & Cornwell, 2016). This is because behavior is directed toward mental representations of the world (rather than its actual state) and these mental representations may not necessarily constitute an accurate rendition of actual circumstances (Mullainathan & Shafir, 2013). Thus, framing helps shape those mental representations.

Information that is vivid and salient usually has a larger impact on behavior than information that is abstract (Sunstein, 2014). In the context of climate change policy, targeted and tailored information provision, in conjunction with wider structural change, is an important policy tool to enable citizens to reduce their carbon dependency and usher in pro-environmental behavior (Lorenzoni, Nicholson-Cole, & Whitmarsh, 2007).

Research has demonstrated that differences in the way environmental issues (climate change competence, engagement, mitigative behavioral intentions) are framed affects individuals' engagement with the issue (Gifford & Comeau, 2011). Large-scale stated choice experiments about ecosystem changes confirm that the same information when presented in a different format leads to different preference parameter estimates (Hoehn, Lupi, & Kaplowitz, 2010).

In a similar vein it has been found that depending on how arguments related to carbon offsetting are framed in terms of economic efficiency, effectiveness, and ethicality, public support for it is affected (Anderson & Bernauer, 2016). Even a seemingly minor change in terminology such as the use of the term "carbon tax" instead of "carbon offset" is found to have a strong influence on the level of support and preference for a certain policy (Hardisty, Johnson, & Weber, 2010).

Studies relating to the framing of fuel efficiency information have found it to be associated with more fuel-efficient choices (Camilleri & Larrick, 2013). In an online experiment and survey, conducted over 10 EU countries, framing energy efficiency information was found to lead to a higher uptake of energy efficient appliances (Leenheer et al., 2014). Similarly, studies looking at the framing of food sustainability information with regard to food purchases (Elsen, Giesen, & Leenheer, 2015) found framing to be associated with pro-environmental behavior.

Framing has been one of the behavioral interventions used by the Regulatory Authority for Electricity, Gas, and Water (AEEGSI) in Italy, to re-design the layout of electricity and utility bills so as to improve energy efficiency (OECD, 2017A). It is also being tested at the Israeli Ministry of Environmental Protection in the context of energy efficiency labels for domestic appliances (OECD, 2017B).

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Nudges as Climate Policy Tools: Some Stylized Facts

Four points may be made with regard to the behavioral tools commonly used in climate change policy. The first is that they are closely related, so, for instance, a default may be viewed as another way of “framing” options. In a similar vein, the provision of information is tied to the manner in which it is presented, making it hard to categorize it as the use of either the information tool or framing tool alone.

Second, some behavioral policy tools are found to be especially effective when used in combination with other behavioral nudges as opposed to being used independently. For instance, an online survey experiment in Netherlands found that social norms when used in conjunction with persuasive information were the most effective in reducing the intent to purchase bottled water (see OFGEM, 2011; van der Linden, 2015).

Third, there is evidence that certain nudges are found to be more effective for certain sections of the target group. For example Datta et al. (2015) in the context of a water savings RCT found social norms to be the most effective behavioral tool for high water-consuming households, whereas goal-setting was found to be the most effective behavioral tool for low-consumption households. The existence of multiple actors as policy targets within the same environment suggests a role for targeted behavioral interventions that are tailored to be most effective based on the specific subgroup in the target population.

Fourth, although the use of nudges as behavioral climate policy instruments has focused on altering individual behavior, nudges are relevant at organizational, national, and international levels as well. For instance, goal-setting can be regarded as the underlying rationale behind getting states to commit to Nationally Determined Contributions as part of the 2015 Paris Conference of the Parties (Tobin et al., 2018). Similarly, social norms with respect to the passage of climate policy legislation elsewhere are seen to influence a state’s behavior in adopting its own climate legislation (Fankhauser, Gennaioli, & Collins, 2016).

While behavioral policy tools are an important application of behavioral science in climate policy, the role of behavioral science is not restricted to this alone. Behavioral science also provides a way of understanding barriers to compliance and sheds light on the impact of behavioral processes of different actors through the policy cycle. This in turn can play an important role in crafting and implementing effective climate change policy.

The Behavioral Barriers Faced by Climate Policy

The underlying assumption of the section on behavioral policy tools was the idea that the desire to comply with policy, in order to behave pro-environmentally, exists among policy-takers and what prevents them from complying are limitations imposed by their own cognitive biases, limited rationality, and behavioral tendencies. Thus, climate policy–relevant

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nudges help mitigate these limitations, often by exploiting these very cognitive biases and enabling behavior to shift in the direction of policy compliance.

However, policy-takers may not always want to comply with policy. They may have varied resources and different capabilities and attitudes when it comes to determining whether they will comply, and—if they do—how and to what extent it will be (Weaver, 2009). Support for various climate policy tools may also be affected by an individual's level of interpersonal and institutional trust (Matti, 2015).

Barriers to Compliance Among Policy-Takers

According to a research study carried out among members of the U.K. public, individual barriers to engaging with climate change include lack of knowledge, uncertainty and skepticism, distrust in information sources, perception of climate change as a distant threat, externalization of blame and responsibility, importance of other priorities, reliance on technology, fatalism and helplessness, and a reluctance to change lifestyles (Lorenzoni et al., 2007). Understanding the barriers to compliance imposed by behavioral processes is important if effective climate policy is to be designed.

First, in general, grasping climate change is challenging because it requires understanding complex aspects of specialized fields (World Bank, 2015); moreover, even science does not have a perfect and complete knowledge of climate change processes—about how climate sensitivities may vary with time and under different scenarios (Clarke, 2010). As a result, people's perception of climate change and its risks acts as a significant barrier to compliance with climate change policy and in behaving in a pro-environment manner.

Second, perceptions of climate change are affected by the cognitive biases and the mental models that people use in relation to climate change. Given the complexity of the issue, individuals often employ the "availability heuristic" (Marx & Weber, 2011). Heuristics, as mentioned, are mental shortcuts that individuals use to reduce the cognitive burden associated with making decisions (Shah & Oppenheimer, 2008). An individual is said to employ an availability heuristic whenever she or he estimates probability or frequency by the ease with which instances or associations related to it can be brought to mind (Tversky & Kahneman, 1973). In the context of climate change, the availability heuristic leads people to believe that the future will be similar to what they have experienced so far (Sunstein, 2006). Employing the availability heuristic on extreme weather conditions that also serve as "focusing events" (Birkland, 2006) affects both individual adaptive decisions and climate policymaking (Juhola, Peltonen, & Niemi, 2012).

Similar to the availability heuristic, is *recency weighting*—the tendency for individuals to overreact to a statistically rare event if that event occurs in the recent past (Zaval & Cornwell, 2016). The availability heuristic and recency weighting result in individual perception of climate change being affected by local weather patterns. So, for instance, Zaval, Keenan, Johnson, and Weber (2014) find that perceptions of a higher than usual local temperature for that day result in an overestimation of warm days through the year, leading to an increased belief in the existence of global warming. Numerous other studies

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confirm this correlation between local weather patterns and beliefs regarding global warming (Krosnick, Holbrook, Lowe, & Visser, 2006; Li, Johnson, & Zaval, 2011; Ungar, 1992).

Another behavioral pattern that people exhibit is reliance on personal experience. Consequently, personal experience of climate change manifestation is found to increase belief in climate change (Spence, Poortinga, Butler, & Pidgeon, 2011). A study among farmers in England found personal experience of flooding to be significantly associated with heightened concerns related to climate change (Hamilton-Webb, Manning, Naylor, & Conway, 2017).

Third, an individual's perceptions of climate change risks may also be tied to his or her larger worldview as well as to cognitive processes by which he or she interprets probabilities (Weber, 2016). The former implies that factors such as ideology and communication of climate change information affect an individual's attitude toward compliance and pro-environmental behavior. For instance, a study in the United States found politically conservative individuals to be less likely to purchase an environmentally friendly product when it was labeled as such, as opposed to when it was unlabeled (Gromet, Kunreuther, & Larrick, 2013). Similarly, support for policies that address climate change increases dramatically when human action is cited as the cause of climate change (Pew Research Center, 2009).

Cognitive processes lead people to underestimate the likelihood of high-probability events and overestimate the likelihood of low-probability events (Kahneman & Tversky, 1979). A study in Mozambique after the occurrence of floods in 2000 found that policymakers had a propensity to overestimate climate-related risks relative to farmers, who in turn preferred the status quo in the face of uncertainty and ambiguity (Patt & Schröter, 2008).

Fourth, a final behavioral barrier to compliance may stem from the perceived temporal and social distance of the consequences of climate change (Weber & Stern, 2011), which in turn is rooted in the "present bias." This cognitive myopia results in people overvaluing immediate benefits and discounting delayed consequences (Frederick, Loewenstein, & O'Donoghue, 2002). A corollary of this is a reluctance to engage in pro-environmental behavior especially if it involves monetary or other non-pecuniary costs.

Thus, behavioral processes play an important role in the policy-taker's perception and engagement with climate change, a manifestation of which are behavioral barriers that hinder compliance with climate change policy and prevent pro-environmental behavior. But behavioral processes also have a role to play in other ways in the policy cycle—through their action on policymakers.

Barriers to Policy Design Among Policymakers

Behavioral processes that affect individual decision-making are not the sole preserve of policy-takers. At the end of the day regulators and policymakers are behavioral agents themselves, and hence not immune to the psychological biases that affect others (Viscusi

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& Gayer, 2015). Lodge and Wegrich (2016) term this the “rationality paradox,” which lies at the heart of the nudging approach. Bounded rationality, they argue, affects policymakers too, whether they are employing behavioral policy tools (which in turn are premised on the logic of bounded rationality on the part of the policy-taker) or traditional policy instruments. Taken to its logical conclusion this implies that behavioral processes of the actors involved come into play at each of the stages of the policy cycle—agenda setting, policy formulation, decision-making, implementation, and evaluation (Howlett, Ramesh, & Perl, 2009).

In the context of climate change policy it implies that the manner in which the problem is framed (for instance as an anthropogenic issue or rooted in natural causes), the solutions that are proposed (for example using traditional policy tools such as a carbon tax or employing behavioral policy tools such as the green default), the decision that is taken, and the way it is implemented and evaluated, are all affected by the behavioral biases of the actors involved at each stage. Five key aspects of behavioral processes related to policymakers are worth highlighting.

First, while many of the cognitive biases associated with policy-takers’ behavior are similar to those faced by policymakers, certain others are more frequently identified with policy-makers. For instance policymakers, like policy-takers, exhibit biased preferences to risk (overestimating small risks and underestimating large risks) (Patt & Schröter, 2008). But the “confirmation bias,” which gives undue weight to certain information or the selective gathering of evidence in order to support a previously held belief (Nickerson, 1998), is more frequently identified with policymakers. This bias presents itself especially in the selection of policy advice. “Sunk cost bias” is yet another bias that marks the behavior of policymakers. It is the tendency to continue with a project once an initial investment of resources has been made (Arkes & Blumer, 1985; World Bank, 2015).

Second, policymakers also employ heuristics, in order to reduce the cognitive load associated with decision-making particularly in complex and uncertain situations. According to the poli-heuristic theory of political decision-making, policymakers employ a two-stage decision process, such that in the first stage they screen available alternatives through cognitive based heuristic techniques (Mintz, 2004). Using a non-compensatory decision rule, politically unacceptable alternatives are eliminated by decision-makers at this point (Dacey & Carlson, 2004). In the second stage with the decision matrix reduced to a smaller number of alternatives, an analytic (i.e., rational) comparison, which typically uses a maximization or lexicographic decision rule, is made to arrive at a final choice (Mintz, 1993, 2004; Mintz, Geva, Redd, & Carnes, 1997).

Third, relevant to policy analysis, is the use of mental models (Hendrick, 1994) by policymakers, which may differ considerably from the mental models that policy-takers employ (World Bank, 2010). In the context of climate change, mental models can shape preferences for different policy options.

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A cross-country study set in Austria, Bangladesh, Finland, Germany, Norway, and the United States found that perceived risk characteristics and mental models of climate change affected support for different climate change policy alternatives. Causal thinking correlated with support for different mitigative policy actions, many of which were not necessarily the most effective (Bostrom et al., 2012). Differences in preferences for climate change policy alternatives may end up dictating actual policy choice as well, especially when the dominant social mental model is in favor of a particular policy alternative, regardless of whether or not it is the most effective one.

Fourth, in addition to the use of mental models, social construction of the target population also exerts an influence on policy design (Schneider & Ingram, 1993). The expected behavior of policy-takers is often framed in terms of positive or negative stereotypes and whether targets are powerful or weak actors. Correspondingly, policy tools tend to be deployed based on such categorizations of the target population (Schneider & Ingram, 1990, 1997).

Such socially reinforced choices and shared mental models can block choices, prevent even conceiving of certain courses of action, and promote certain other actions (World Bank, 2010). Thus, designing for a different set of motivations of policy targets and linking these to specific tool choices is complex. It is thus prudent for policymakers to understand behavioral aspects of policy tools and policy targets, to ensure that these match in order to achieve an effective compliance regime (Howlett, 2016).

Finally, behavioral failure at the level of the individual policymaker may be reflected at the level of the government through the policies that come into being. For instance, biased preferences to risk tend to be embodied in larger governmental policies (Viscusi & Gayer, 2015). Similarly, “loss aversion” (the tendency of overvaluing losses relative to gains) tends to also characterize government policies (McDermott, 1992; Whyte & Levi, 1994).

Thus, behavioral processes exert a strong influence through the policy cycle on both the policy-taker and policymaker. Understanding this and crafting policy accordingly is important to counter climate change. The approach is not without its challenges, however.

Challenges to the Use of Behavioral Sciences in Climate Policy

Although behavioral sciences provide a policy toolkit as well as a way of understanding barriers to compliance and policy design in climate policy, they still have a long way to go before this understanding of individual behavioral processes can be translated into comprehensive and sustainable policy options. More research is needed to understand what works, in what context, and to what extent (John et al., 2013).

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In particular, behavioral sciences have tended to ignore the role of cultural differences. Levinson and Peng (2007) argue that much of the evidence on behavioral insights is tailored to Western settings and affected by culturally guided assumptions about human minds, desires, and rationality. For instance, in the case of framing effects they find that judgements of financial value and property ownership differ dramatically between Chinese and American study participants. This then represents a gap that is relevant to behaviorally premised climate policy as well.

In particular, behavioral processes that are rooted in mental models or affected by social constructions of the policy target group may be hard to change. Such adjustment, if and when it does happen, may be too late for policies aiming to forestall climate disruption, given the inertia of the climate system (World Bank, 2015).

Moreover, infrastructural contexts (Vliet, Chappells, & Shove, 2005), as well as perceptions of helplessness (Lertzman, 2015), are significant barriers towards low carbon behavior change (Büchs et al., 2018) that are difficult to overcome solely through behaviorally premised insights and policy tools.

Thus, it is unlikely that behavioral insights would completely supplant traditional climate policy instruments any time in the near future. While behavioral sciences have the potential to bring newer policy tools into the climate policy toolkit, these will need to be supported with the continued use of traditional policy tools (Moseley & Stoker, 2013).

An oft directed criticism of the nudging approach is its inability to usher in long-lasting behavioral modifications (Mills, 2013) because of its overemphasis on individual preferences and reliance on an atomistic approach to social structure (Goodwin, 2012). For climate change action, where sustained behavioral modifications are especially necessary, this is particularly relevant. One of the few long-running energy conservation studies in the United States, finds, even after two years, evidence of long-term impacts (Allcott & Rogers, 2014), but a general paucity of research studies that track behavioral changes over long periods of time makes it difficult to conclusively establish the temporal stability (or instability) of shifts in behavior.

Related to this is the problem of “placation,” wherein a nudge successfully modifies behavior in the short term, but because the long-term problem remains unaddressed, it leads to the problem exacerbating over time (Lodge & Wegrich, 2016). For instance, a mitigative nudge toward using biofuels instead of liquid fossil fuels can in the long run have negative implications for adaptation through its potentially adverse effect on food production and security, as biofuel production replaces more diverse ecosystems (Moser, 2012).

Nudges may also have an unintended “behavioral spillover” (Dolan & Galizzi, 2015). An example of this is the rebound effect in the energy sector. In the case of energy efficiency, the rebound effect describes how consumers increase product usage as their energy costs decrease (Miller & Mannix, 2016). Borenstein (2013) through illustrative calculations es-

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timates that the rebound effect significantly reduces net savings from energy efficiency improvements.

Finally, nudges have been criticized across the board (not just in the context of climate policy) on ethical grounds in that they violate an individual's right to make choices in his or her own interest as the person understands them (Goodwin, 2012; White, 2016). There are also other concerns that nudging runs the risk of infantilizing and diminishing people's autonomous decision-making capacities (Bovens, 2009; Hausman & Welch, 2010).

Tied to these larger concerns are challenges specific to the commonly used nudges in climate change policy. For the information tool and framing tools to be effective there needs to be high levels of trust and credibility associated with the information that is provided and the information provider (Howlett, 2016). In general, interpersonal and institutional trust are key ingredients for public support for climate policy tools (Matti, 2015). But if nudging leads to a general distrust of the government, it may well undermine these efforts (Mols, Haslam, Jetten, & Steffens, 2015; Wilkinson, 2013).

Moreover, citizens are exposed to many competing claims ("frames" and "counter frames") about climate change and climate policy measures. This information abundance means framing effects for individuals may be lessened because of prior knowledge, attitude, and interest in climate issues, with the exception of those who are ambivalent toward the issue or know little about it (Bernauer & McGrath, 2016).

Attention also needs to be paid to identifying the appropriate social norm and highlighting it. There is evidence that social norms may inadvertently normalize undesirable behavior by drawing attention to how widespread it is (Cialdini, 2003). There is also the danger that an emphasis on social norms could trigger resistance, especially among younger people (Sunstein & Reisch, 2016) and "deviant subcommunities" (Kagan & Skolnick, 1993).

For goal-setting it has been found that it is important for the goal to be realistic in order for it to be successful (Harding & Hsiaw, 2014). But selecting the correct default goals or enabling policy targets to commit to an appropriate goal is not easy.

Finally, in the case of green defaults, an underlying challenge is determining the appropriate default. Sensible defaults may be hard to establish because of the many considerations people take into account when making decisions. There may be distributional issues associated with a default if it imposes net costs on all targets (including the economically disadvantaged). In this context it has been found that defaults tend to be stickier for low-income individuals, perhaps due to, what Mullainathan & Shafir (2014) argue is the cognitive burden imposed by scarcity. This suggests that green defaults are likely to have a regressive impact. Finally, defaults may not work if individuals have clear preferences (Sunstein & Reisch, 2016). Based on an experimental study set in Switzerland, Ghesla (2017) found that green electricity defaults are unable to match people's preferences.

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Thus, the oft-assumed implicitness of defaults matching preferences needs to be studied further.

Conclusion: Moving Forward With Behavioral Tools

Behavioral sciences are making a large impact across the world in public policy, regulation, and law (Sunstein, 2016). They have provided a valuable toolbox to enrich understanding of human decision-making, and thereby allow for improvement in the design of public policy on a context-specific case-by-case basis (Oliver, 2013). In the case of climate change policy, this has taken the form of five commonly used behavioral nudges: provision of information, goal-setting, defaults, social norms, and framing. These policy tools have been used to increase pro-environmental behavior and help policy-takers comply with the goals of climate policy.

The evidence on climate relevant nudges has generated some important lessons. First, a combination of nudges, rather than a single nudge, may be more effective in ushering in desired behavior. Second, nudges in conjunction with traditional policy instruments will be required for effective climate policy. Third, nudges may be most effective if they can be tailored to specific groups. One way to do this is by examining barriers to change for a specific target group and tailoring interventions to address those barriers (see McKenzie-Mohr, 2000).

Although climate-relevant nudges have been used successfully at the level of the individual, the mechanism underlying certain behavioral nudges (like social norms and goal-setting) is also seen in climate policies that are operating at more macro levels. This suggests a direction for future research in scaling up other individual-level nudges to higher levels.

Despite the criticisms directed at behavioral interventions, there remains considerable public support for their use. In particular, Reisch and Sunstein (2016), based on nationally representative surveys in six European nations (Denmark, France, Germany, Hungary, Italy, and the United Kingdom), found that the majority supported default rules encouraging and mandating green energy use. Similarly, Hagman et al. (2015) found a high degree of acceptance for “nudging” policies among a sample drawn from the United States and Sweden. Within this sample, however, the policies that the researchers classified as pro-social (i.e., focusing on social welfare) had a significantly lower acceptance rate than those considered pro-self-nudge (i.e., focusing on private welfare).

In case of climate change policy, while behavioral tools by themselves seem unlikely to provide the magic bullet to overhaul individual behavior in line with climate change goals, they offer innovative alternatives for mitigation and adaptation efforts.

Given that people’s attitudes about climate change are malleable, systematic biases can have a powerful influence (Zaval & Cornwell, 2016). Thus, an important role of behavioral science in climate policy lies in addressing issues of public perception of climate change

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and in facilitating the creation of a more responsive energy demand, capable of adapting to changes in renewable energy supply (Pollitt & Shaorshadze, 2011).

Mills (2013) argues that such techniques have a valid role to play in solving large-scale policy issues, but only in tandem with other policy options. Behavioral tools can complement economic tools in climate change policy, rather than substituting for them. Behavioral policy tools will be most effective when they are supported by larger government policies and are consistent with them (Zaval & Cornwell, 2016) and when the value orientation frame of the policy-takers matches the policy (Berglund & Matti, 2006).[10]

In this context another important role for behavioral sciences is to identify ways to convince targets to support, and policymakers to adopt, what are known to be effective economic tools (for example carbon pricing, cap and trade) to curb greenhouse gas emissions (World Bank, 2010).

Research studies thus far on non-economic behavioral interventions in the domain of climate change have as a general rule been limited in size, with interventions not monitored for prolonged periods of time, thereby making it difficult to determine whether behavior and habits persist or eventually return to pre-intervention norms (Pollitt & Shaorshadze, 2011). The way forward would thus be in the direction of continued monitoring to identify short-term and long-term effects, as well as any unintended consequences and behavioral spillovers of behavioral interventions (OECD, 2017A).

Finally, the key to applying behavioral insights in climate change policy will be reliable and robust data that stands up to public scrutiny. While RCTs offer some of the highest internal validity in research studies (Shadish, Cook, & Campbell, 2002), especially for policy evaluation, they need to be complemented with methods that help build theoretical and conceptual knowledge so that questions beyond “what works?” to “why does it work?” can be answered (Deaton & Cartwright, 2016). Other methodologies (such as quasi-experiments, qualitative research, and surveys) may complement research on behavioral interventions, particularly at the stage of problem diagnosis to provide a better understanding of the issue (Sousa Lourenço et al., 2016). However, where evidence is lacking, there is a role for policy pilots to discern behavioral responses by testing the assumptions of the policy responses of targets on a small scale before a roll out the complete program (Nair & Howlett, 2015; Vreugdenhil, Taljaard, & Slinger, 2012).

References

Abrahamse, W., Steg, L., Vlek, C., & Rothengatter, T. (2007). **The effect of tailored information, goal setting, and tailored feedback on household energy use, energy-related behaviors, and behavioral antecedents**. *Journal of Environmental Psychology*, *27*(4), 265–276.

Allan, J., Querstret, D., Banas, K., & de Bruin, M. (2017). **Environmental interventions for altering eating behaviours of employees in the workplace: A systematic review**. *Obesity Reviews*, *18*(2), 214–226.

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date: 30 July 2019

Allcott, H. (2011). **Social norms and energy conservation**. *Journal of Public Economics*, *95*(9–10), 1082–1095.

Allcott, H., & Rogers, T. (2014). **The short-run and long-run effects of behavioral interventions: Experimental evidence from energy conservation**. *American Economic Review*, *104*(10), 3003–3037.

Anderson, B., & Bernauer, T. (2016). **How much carbon offsetting and where? Implications of efficiency, effectiveness, and ethicality considerations for public opinion formation**. *Energy Policy*, *94*, 387–395.

Araña, J. E., & León, C. J. (2013). **Can defaults save the climate? Evidence from a field experiment on carbon offsetting programs**. *Environmental and Resource Economics*, *54*(4), 613–626.

Arkes, H. R., & Blumer, C. (1985). **The psychology of sunk cost**. *Organizational Behavior and Human Decision Processes*, *35*(1), 124–140.

Ayres, I., Raseman, S., & Shih, A. (2013). **Evidence from two large field experiments that peer comparison feedback can reduce residential energy usage**. *Journal of Law, Economics, and Organization*, *29*(5), 992–1022.

Baca-Motes, K., Brown, A., Gneezy, A., Keenan, E. A., & Nelson, L. D. (2013). **Commitment and behavior change: Evidence from the field**. *Journal of Consumer Research*, *39*(5), 1070–1084.

Banerjee, A., & Duflo, E. (2011). *Poor economics: A radical rethinking of the way to fight global poverty* (Reprint ed.). New York, NY: Public Affairs.

Barr, M. S., Mullainathan, S., & Shafir, E. (2009). The case for behaviorally informed regulation. In D. Moss & J. Cisternino (Eds.), *New perspectives on regulation* (pp. 25–62). Cambridge, MA: The Tobin Project.

Bavel, R., van, Herrmann, B., Esposito, G., Proestakis, A., & Institute for Prospective Technological Studies. (2013). *Applying behavioural sciences to EU policy-making*. Luxembourg, Belgium: Publications Office of the European Union.

Behavioural Economics Team of the Australian Government. (2016, April 22). **Behavioural Economics Team of the Australian Government. Department of the Prime Minister and Cabinet**.

Behavioural Insights Team. (2010). ***Applying behavioural insight to health***. London, UK: Cabinet Office Behavioural Insights Team.

Behavioural Insights Team. (2011). ***Behaviour change and energy use***. London, UK: Cabinet Office Behavioural InsightsTeam.

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date: 30 July 2019

Behavioural Insights Team. (2014). ***EAST: Four simple ways to apply behavioural insights***. London, UK: Cabinet Office Behavioural Insights Team.

Berglund, C., & Matti, S. (2006). **Citizen and consumer: the dual role of individuals in environmental policy**. *Environmental Politics*, *15*(4), 550–571.

Bernauer, T., & McGrath, L. F. (2016). **Simple reframing unlikely to boost public support for climate policy**. *Nature Climate Change*, *6*(7), 680–683.

Biermann, F., Kanie, N., & Kim, R. E. (2017). **Global governance by goal-setting: The novel approach of the UN Sustainable Development Goals**. *Current Opinion in Environmental Sustainability*, *26–27*, 26–31.

Birkland, T. A. (2006). ***Lessons of disaster: Policy change after catastrophic events***. Washington, DC: Georgetown University Press.

Borenstein, S. (2013). ***A microeconomic framework for evaluating energy efficiency rebound and some implications***. Working Paper No. 19044. National Bureau of Economic Research.

Bostrom, A., O'Connor, R. E., Böhm, G., Hanss, D., Bodi, O., Ekström, F., . . . Sælensminde, I. (2012). **Causal thinking and support for climate change policies: International survey findings**. *Global Environmental Change*, *22*(1), 210–222.

Bovens, L. (2009). **The ethics of nudge**. In T. Grüne-Yanoff & S. O. Hansson (Eds.), *Preference change* (pp. 207–219). Dordrecht, The Netherlands: Springer.

Brick, K., De, S., & Visser, M. M. (2017). ***Behavioural nudges for water conservation: Experimental evidence from Cape Town***.

Brown, C. L., & Krishna, A. (2004). **The skeptical shopper: A metacognitive account for the effects of default options on choice**. *Journal of Consumer Research*, *31*(3), 529–539.

Büchs, M., Bahaj, A. S., Blunden, L., Bourikas, L., Falkingham, J., James, P., . . . Wu, Y. (2018). **Promoting low carbon behaviours through personalised information? Long-term evaluation of a carbon calculator interview**. *Energy Policy*, *120*, 284–293.

Buuren, A., van, Driessen, P. P. J., van Rijswick, M., Rietveld, P., Salet, W., Spit, T., & Teisman, G. (2013). **Towards adaptive spatial planning for climate change: Balancing between robustness and flexibility**. *Journal for European Environmental & Planning Law*, *10*(1), 29–53.

Camerer, C. (1999). **Behavioral economics: Reunifying psychology and economics**. *Proceedings of the National Academy of Sciences*, *96*(19), 10575–10577.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Camerer, C. (2004). Prospect theory in the wild: Evidence from the field. In C. Camerer, G. Loewenstein, & M. Rabin (Eds.), *Advances in behavioral economics* (pp. 148–161). Princeton, NJ: Princeton University Press.

Camerer, C., Issacharoff, S., Loewenstein, G., O'Donoghue, T., & Rabin, M. (2003). Regulation for conservatives: Behavioral economics and the case for "asymmetric paternalism. *University of Pennsylvania Law Review*, *151*(3), 1211–1254.

Camilleri, A. R., & Larrick, R. P. (2013). **Metric and scale design as choice architecture tools**. *Journal of Public Policy & Marketing*, *33*(1), 108–125.

Centre d'Analyse Stratégique. (2011). ***Green nudges: New incentives for ecological behavior***. Paris, France: Premier Ministre Republique Francaise.

Centre for Sustainable Energy and the Environmental Change Institute, University of Oxford. (2012). ***What are the factors influencing energy behaviours and decision-making in the non-domestic sector? A rapid evidence assessment***.Policy paper. Department of Energy & Climate Change, UK Government.

Cialdini, R. B. (2003). Crafting normative messages to protect the environment. *Current Directions in Psychological Science*, *12*(4), 105–109.

Cialdini, R. B. (2008). *Influence: Science and practice* (5th ed.). Boston, MA: Allyn and Bacon.

Clarke, H. (2010). Formulating policy responses to global warming in the face of uncertainty. *Agenda: A Journal of Policy Analysis and Reform*, *17*(1), 33–54.

Codagnone, C., Bogliacino, F., & Veltri, G. (2013). ***Testing CO2/Car labelling options and consumer information***. No. Final Report. Brussels, Belgium: European Commission.

Dacey, R., & Carlson, L. J. (2004). Traditional decision analysis and the poliheuristic theory of foreign policy decision making. *Journal of Conflict Resolution*, *48*(1), 38–55.

Datta, S., Miranda, J. J., Zorattoa, L., Calvo-González, O., Darling, M., & Lorenzana, K. (2015). ***A behavioral approach to water conservation: Evidence from Costa Rica***. The World Bank.

Datta, S., & Mullainathan, S. (2014). **Behavioral design: A new approach to development policy**. *Review of Income and Wealth*, *60*(1), 7–35.

Deaton, A., & Cartwright, N. (2016). ***Understanding and misunderstanding randomized controlled trials*** No. w22595. Cambridge, MA: National Bureau of Economic Research.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Delmas, M. A., Fischlein, M., & Asensio, O. I. (2013). **Information strategies and energy conservation behavior: A meta-analysis of experimental studies from 1975 to 2012**. *Energy Policy*, *61*, 729–739.

Department of Energy and Climate Change & Behavioural Insights Team. (2014). ***Evaluation of the DECC and John Lewis energy labelling trial***. Research and analysis No. 14D/320. Department of Energy & Climate Change, UK Government.

Dinner, I., Johnson, E. J., Goldstein, D. G., & Liu, K. (2011). **Partitioning default effects: Why people choose not to choose**. *Journal of Experimental Psychology: Applied*, *17*(4), 332–341.

Dolan, P., & Galizzi, M. M. (2014). **Getting policy-makers to listen to field experiments**. *Oxford Review of Economic Policy*, *30*(4), 725–752.

Dolan, P., & Galizzi, M. M. (2015). **Like ripples on a pond: Behavioral spillovers and their implications for research and policy**. *Journal of Economic Psychology*, *47*, 1–16.

Dolan, P., Hallsworth, M., Halpern, D., King, D., & Vlaev, I. (2010). ***Mindspace: Influencing behaviour through public policy***. London, UK: Institute for Government.

Dolan, P., & Metcalfe, R. (2013). ***Neighbors, knowledge, and nuggets: two natural field experiments on the role of incentives on energy conservation***. CEP Discussion Papers No. CEPDP1222. London, UK: Centre for Economic Performance, London School of Economics and Political Science.

Duflo, E., Kremer, M., & Robinson, J. (2011). **Nudging farmers to use fertilizer: Theory and experimental evidence from Kenya**. *American Economic Review*, *101*(6), 2350–2390.

Ebeling, F., & Lotz, S. (2015). **Domestic uptake of green energy promoted by opt-out tariffs**. *Nature Climate Change*, *5*(9), 868–871.

Elsen, M., Giesen, R., & Leenheer, J. (2015). ***Milan BExpo 2015: Behavioural study on food choices and eating habits: Final report***. Luxembourg, Belgium: Publications Office of the European Union.

Elster, J. (1989). *The cement of society: A study of social order*. Cambridge, UK: Cambridge University Press.

Environmental Protection Agency. (2001). *The United States experience with economic incentives for protecting the environment*. No. EPA-240-R-01-001. National Center for Environmental Economics. Office of Policy, Economics, and Innovation Office of the Administrator U.S. Environmental Protection Agency.

Fankhauser, S., Gennaioli, C., & Collins, M. (2016). **Do international factors influence the passage of climate change legislation?** *Climate Policy*, *16*(3), 318–331.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Fehr, E., & Schmidt, K. M. (2003). **Theories of fairness and reciprocity: Evidence and economic applications**. In M. Dewatripont, L. P. Hansen, & S. J. Turnovsky (Eds.), *Advances in economics and econometrics* (pp. 208–257). Cambridge, UK: Cambridge University Press.

Ferraro, P. J., Miranda, J. J., & Price, M. K. (2011). **The persistence of treatment effects with norm-based policy instruments: Evidence from a randomized environmental policy experiment**. *American Economic Review*, *101*(3), 318–322.

Ferraro, P. J., & Price, M. K. (2013). **Using nonpecuniary strategies to influence behavior: Evidence from a large-scale field experiment**. *Review of Economics and Statistics*, *95*(1), 64–73.

Festinger, L. (1962). *A theory of cognitive dissonance*. Stanford, CA: Stanford University Press.

Fleig, A., Schmidt, N. M., & Tosun, J. (2017). **Legislative dynamics of mitigation and Adaptation Framework Policies in the EU**. *European Policy Analysis*, *3*(1), 101–124.

Frederick, S., Loewenstein, G., & O'Donoghue, T. (2002). Time discounting and time preference: A critical review. *Journal of Economic Literature*, *40*(2), 351–401.

Frederiks, E. R., Stenner, K., & Hobman, E. V. (2015). **Household energy use: Applying behavioural economics to understand consumer decision-making and behaviour**. *Renewable and Sustainable Energy Reviews*, *41*, 1385–1394.

Fujii, S., & Taniguchi, A. (2005). **Reducing family car-use by providing travel advice or requesting behavioral plans: An experimental analysis of travel feedback programs**. *Transportation Research Part D: Transport and Environment*, *10*(5), 385–393.

Galizzi, M. M. (2017). Behavioral aspects of policy formulation: Experiments, behavioral insights, nudges. In M. Howlett, I. Mukherjee, & S. Fraser (Eds.), *Handbook of policy formulation* (p. 410). Cheltenham, UK: Edward Elgar.

Gerard, D., & Lave, L. (2007). **Experiments in technology forcing: Comparing the regulatory processes of US automobile safety and emissions regulations**. *International Journal of Technology, Policy and Management*, *7*(1), 1.

Ghesla, C. (2017). **Defaults in green electricity markets: Preference match not guaranteed**. *Journal of the Association of Environmental and Resource Economists*, *4*(S1), S37–S84.

Gifford, R., & Comeau, L. A. (2011). **Message framing influences perceived climate change competence, engagement, and behavioral intentions**. *Global Environmental Change*, *21*(4), 1301–1307.

Glennerster, R., & Takavarasha, K. (2013). ***Running randomized evaluations: A practical guide***.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Goldstein, N. J., Cialdini, R. B., & Griskevicius, V. (2008). **A room with a viewpoint: Using social norms to motivate environmental conservation in hotels**. *Journal of Consumer Research*, *35*(3), 472–482.

Goodwin, T. (2012). **Why we should reject "nudge"**. *Politics*, *32*(2), 85–92.

Goulder, L. H., & Schein, A. R. (2013). **Carbon taxes versus cap and trade: A critical review**. *Climate Change Economics*, *4*(3), 1350010.

Gromet, D. M., Kunreuther, H., & Larrick, R. P. (2013). **Political ideology affects energy-efficiency attitudes and choices**. *Proceedings of the National Academy of Sciences*, *110*(23), 9314–9319.

Hagman, W., Andersson, D., Västfjäll, D., & Tinghög, G. (2015). **Public views on policies involving nudges**. *Review of Philosophy and Psychology*, *6*(3), 439–453.

Hallsworth, M. (2016, August 1). ***Behavioural insights—The road to "better" policies?***

Halpern, D. (2015). *Inside the nudge unit: How small changes can make a big difference*. London, UK: W. H. Allen.

Hamilton-Webb, A., Manning, L., Naylor, R., & Conway, J. (2017). **The relationship between risk experience and risk response: A study of farmers and climate change**. *Journal of Risk Research*, *20*(11), 1379–1393.

Harding, M., & Hsiaw, A. (2014). **Goal setting and energy conservation**. *Journal of Economic Behavior & Organization*, *107*, 209–227.

Hardisty, D. J., Johnson, E. J., & Weber, E. U. (2010). **A dirty word or a dirty world?: Attribute framing, political affiliation, and query theory**. *Psychological Science*, *21*(1), 86–92.

Hargadon, A. B., & Douglas, Y. (2001). **When innovations meet institutions: Edison and the design of the electric light**. *Administrative Science Quarterly*, *46*(3), 476.

Harrison, D., Foss, A., Klevnas, P., & Radov, D. (2011). ***Economic policy instruments for reducing greenhouse gas emissions***.

Hausman, D. M., & Welch, B. (2010). **Debate: To nudge or not to nudge**. *Journal of Political Philosophy*, *18*(1), 123–136.

Head, B. W., & Alford, J. (2015). **Wicked problems: Implications for public policy and management**. *Administration & Society*, *47*(6), 711–739.

Hendrick, R. (1994). **A heuristic approach to policy analysis and the use of sensitivity analysis**. *Public Productivity & Management Review*, *18*(1), 37–55.

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date: 30 July 2019

Hoehn, J. P., Lupi, F., & Kaplowitz, M. D. (2010). Stated choice experiments with complex ecosystem changes: The effect of information formats on estimated variances and choice parameters. *Journal of Agricultural and Resource Economics*, *35*(3), 568–590.

Hollands, G. J., Shemilt, I., Marteau, T. M., Jebb, S. A., Kelly, M. P., Nakamura, R., . . . Ogilvie, D. (2013). **Altering micro-environments to change population health behaviour: Towards an evidence base for choice architecture interventions**. *BMC Public Health*, *13*, 1218.

Hood, C. (1986). *The tools of government*. Chatham, NJ: Chatham House.

Howlett, M. (2016). **Policy tools & their targets: Beyond nudges and utility maximization in policy compliance**. In *Workshop No. 2: Behavioural Change and Public Policy*. Pisa, Italy.

Howlett, M., Ramesh, M., & Perl, A. (2009). *Studying public policy: Policy cycles and policy subsystems* (3rd ed.). New York, NY: Oxford University Press.

Intergovernmental Panel on Climate Change. (2001). Working Group III: Mitigation. In R. T. Watson & Core Writing Team (Eds.), *Climate change 2001—IPCC Third Assessment Report*. Cambridge, UK: Cambridge University Press.

Intergovernmental Panel on Climate Change. (2014a). Annex II: Glossary (K. J. Mach, S. Planton, & C. von Stechow, Eds.). In R. K. Pachauri, L. Mayer, & Intergovernmental Panel on Climate Change (Eds.), *Climate change 2014: Synthesis report: Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (pp. 117–130). Geneva, Switzerland: Intergovernmental Panel on Climate Change.

Intergovernmental Panel on Climate Change. (2014b). *Climate change 2014: Impacts, adaptation, and vulnerability: Working Group II contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (C. B. Field & V. R. Barros, Eds.). New York, NY: Cambridge University Press.

Intergovernmental Panel on Climate Change. (2014c). *Climate change 2014: Mitigation of climate change: Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (O. Edenhofer, R. Pichs-Madruga, Y. Sokona, E. Farahani, S. Kadner, K. Seyboth, . . . J. C. Minx, Eds.). New York, NY: Cambridge University Press.

John, P., Cotterill, S., Moseley, A., Richardson, L., Smith, G., Stoker, G., & Wales, C. (2013). *Nudge, nudge, think, think: Experimenting with ways to change civic behaviour* (Rev. paperback ed.). London, UK: Bloomsbury.

Johnson, E. J., & Goldstein, D. G. (2013). Decisions by default. In E. Shafir (Ed.), *The behavioral foundations of public policy* (pp. 417–427). Princeton, NJ: Princeton University Press.

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date: 30 July 2019

Jones, B. D., Boushey, G., & Workman, S. (2006). **Behavioral rationality and the policy processes: Toward a new model of organizational information processing**. In B. G. Peters & J. Pierre (Eds.) *Handbook of public policy* (pp. 49–74). London, UK: SAGE.

Juhola, S., Peltonen, L., & Niemi, P. (2012). **The ability of Nordic countries to adapt to climate change: Assessing adaptive capacity at the regional level**. *Local Environment*, *17*(6–7), 717–734.

Kagan, R. A., & Skolnick, J. H. (1993). Banning smoking: Compliance without enforcement. *Smoking Policy: Law, Politics, and Culture*, *69*, 78–80.

Kahneman, D. (2003). **Maps of bounded rationality: Psychology for behavioral economics**. *American Economic Review*, *93*(5), 1449–1475.

Kahneman, D. (2013). Foreword. In E. Shafir (Ed.), *The behavioral foundations of public policy* (pp. vii–x). Princeton, NJ: Princeton University Press.

Kahneman, D., & Tversky, A. (1979). **Prospect theory: An analysis of decision under risk**. *Econometrica*, *47*(2), 263–291.

Kahneman, D., & Tversky, A. (Eds.). (2000). *Choices, values, and frames*. New York, NY: Cambridge University Press.

Kallbekken, S., Sælen, H., & Hermansen, E. A. T. (2013). **Bridging the energy efficiency gap: A field experiment on lifetime energy costs and household appliances**. *Journal of Consumer Policy*, *36*(1), 1–16.

Karlan, D. S., & Appel, J. (2011). *More than good intentions: How a new economics is helping to solve global poverty*. New York, NY: Dutton.

Keskitalo, E. C. H. (2010). *Developing adaptation policy and practice in Europe: Multi-level governance of climate change*. Dordrecht, The Netherlands: Springer.

Klein, R. J. T., Schipper, E. L. F., & Dessai, S. (2005). **Integrating mitigation and adaptation into climate and development policy: Three research questions**. *Environmental Science & Policy*, *8*(6), 579–588.

Krause, R. M. (2011). **Policy innovation, intergovernmental relations, and the adoption of climate protection initiatives by U.S. cities**. *Journal of Urban Affairs*, *33*(1), 45–60.

Krosnick, J. A., Holbrook, A. L., Lowe, L., & Visser, P. S. (2006). **The origins and consequences of democratic citizens' policy agendas: A study of popular concern about global warming**. *Climatic Change*, *77*(1–2), 7–43.

Kuhfuss, L., Préget, R., Thoyer, S., & Hanley, N. (2016). **Nudging farmers to enrol land into agri-environmental schemes: The role of a collective bonus**. *European Review of Agricultural Economics*, *43*(4), 609–636.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Lechtenböhmer, S., & Luhmann, H.-J. (2013). Decarbonization and regulation of Germany's electricity system after Fukushima. *Climate Policy*, *13*(supp. 1), 146–154.

Leenheer, J., Elsen, M., Mikola, N., Wagt, M., van der, Lloyd, L., European Commission, . . . GFK. (2014). ***Study on the effects on consumer behaviour of online sustainability information displays: final report***. Luxembourg, Belgium: Publications Office of the European Union.

Leggett, J. A. (2011). ***Climate change: Conceptual approaches and policy tools***. CRS Report for Congress No. 7–5700. Congressional Research Service.

Lertzman, R. (2015). *Environmental melancholia: Psychoanalytic dimensions of engagement*. New York, NY: Routledge.

Levin, K., Cashore, B., Bernstein, S., & Auld, G. (2012). **Overcoming the tragedy of super wicked problems: Constraining our future selves to ameliorate global climate change**. *Policy Sciences*, *45*(2), 123–152.

Levinson, J. D., & Peng, K. (2007). **Valuing cultural differences in behavioral economics**. *ICFAI Journal of Behavioral Finance*, *4*, 32–47.

Li, Y., Johnson, E. J., & Zaval, L. (2011). **Local warming: Daily temperature change influences belief in global warming**. *Psychological Science*, *22*(4), 454–459.

Lindblom, C. E. (1959). **The science of "muddling through"**. *Public Administration Review*, *19*(2), 79–88.

Lipscy, P. Y., & Schipper, L. (2013). **Energy efficiency in the Japanese transport sector**. *Energy Policy*, *56*, 248–258.

Locke, E. A., & Latham, G. P. (2002). Building a practically useful theory of goal setting and task motivation: A 35-year odyssey. *American Psychologist*, *57*(9), 705.

Lodge, M., & Wegrich, K. (2016). **The rationality paradox of nudge: Rational tools of government in a world of bounded rationality**. *Law & Policy*, *38*(3), 250–267.

Loewenstein, G. (2008). *Exotic preferences: Behavioral economics and human motivation*. Oxford, UK: Oxford University Press.

Lokhorst, A. M., Werner, C., Staats, H., van Dijk, E., & Gale, J. L. (2013). **Commitment and behavior change: A meta-analysis and critical review of commitment-making strategies in environmental research**. *Environment and Behavior*, *45*(1), 3–34.

Loock, C.-M., Staake, T., & Thiesse, F. (2013). **Motivating energy-efficient behavior with green IS: An investigation of goal setting and the role of defaults**. *MIS Quarterly*, *37*(4), 1313–1332.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Lorenzoni, I., Nicholson-Cole, S., & Whitmarsh, L. (2007). **Barriers perceived to engaging with climate change among the UK public and their policy implications**. *Global Environmental Change*, *17*(3–4), 445–459.

Low, D. (2012). *Behavioural economics and policy design: Examples from Singapore*. Singapore: World Scientific.

Lunn, P. (2013). Behavioural economics and policymaking: Learning from the early adopters. *Economic and Social Review*, *43*(3), 423–449.

Lunn, P. (2014). ***Regulatory policy and behavioural economics***. Paris, France: Organisation for Economic Co-operation and Development.

Marx, S. M., & Weber, E. U. (2011). Decision making under climate uncertainty: The power of understanding judgment and decision processes. In T. Dietz & D. Bidwell (Eds.), *Climate change in the Great Lakes region: Navigating an uncertain future* (pp. 99–128). East Lansing: Michigan State University Press.

Massey, E., Biesbroek, R., Huitema, D., & Jordan, A. (2014). **Climate policy innovation: The adoption and diffusion of adaptation policies across Europe**. *Global Environmental Change*, *29*, 434–443.

Matti, S. (2015). **Chapter 35: Climate policy instruments**. In K. Bäckstrand & E. Lövbrand (Eds.), *Research handbook on climate governance* (pp. 400–410). Cheltenham, UK: Edward Elgar.

McDermott, R. (1992). **Prospect theory in international relations: The Iranian hostage rescue mission**. *Political Psychology*, *13*(2), 237–263.

McKenzie, C. R. M., Liersch, M. J., & Finkelstein, S. R. (2006). **Recommendations implicit in policy defaults**. *Psychological Science*, *17*(5), 414–420.

McKenzie-Mohr, D. (2000). **New ways to promote proenvironmental behavior: Promoting sustainable behavior: An introduction to community-based social marketing**. *Journal of Social Issues*, *56*(3), 543–554.

McKirnan, D. J. (1980). **The identification of deviance: A conceptualization and initial test of a model of social norms**. *European Journal of Social Psychology*, *10*(1), 75–93.

Meloni, I., Spissu, E., & Bhat, C. (2011). The effect of personal cap-and-trade mileage policies on individual activity-travel patterns: The Activity Locator project. *Transportation Letters*, *3*(4), 293–307.

Miller, S. E., & Mannix, B. F. (2016). **One standard to rule them all: The disparate impact of energy efficiency regulations**. In S. Abdukadirov (Ed.), *Nudge theory in action* (pp. 251–287). Cham, Switzerland: Springer International.

PRINTED FROM the OXFORD RESEARCH ENCYCLOPEDIA, CLIMATE SCIENCE (oxfordre.com/climatescience). (c) Oxford University Press USA, 2019. All Rights Reserved. Personal use only; commercial use is strictly prohibited (for details see Privacy Policy and Legal Notice).

date: 30 July 2019

Mills, C. (2013). **Why nudges matter: A reply to Goodwin**. *Politics*, *33*(1), 28–36.

Mintz, A. (1993). The decision to attack Iraq: A noncompensatory theory of decision making. *Journal of Conflict Resolution*, *37*(4), 595–618.

Mintz, A. (2004). How do leaders make decisions? A poliheuristic perspective. *Journal of Conflict Resolution*, *48*(1), 3–13.

Mintz, A., Geva, N., Redd, S. B., & Carnes, A. (1997). **The effect of dynamic and static choice sets on political decision making: An analysis using the decision board platform**. *American Political Science Review*, *91*(3), 553–566.

Mols, F., Haslam, S. A., Jetten, J., & Steffens, N. K. (2015). **Why a nudge is not enough: A social identity critique of governance by stealth**. *European Journal of Political Research*, *54*(1), 81–98.

Moore, D. A., & Healy, P. J. (2008). **The trouble with overconfidence**. *Psychological Review*, *115*(2), 502–517.

Moseley, A., & Stoker, G. (2013). **Nudging citizens? Prospects and pitfalls confronting a new heuristic**. *Resources, Conservation and Recycling*, *79*, 4–10.

Moser, S. C. (2012). **Adaptation, mitigation, and their disharmonious discontents: An essay**. *Climatic Change*, *111*(2), 165–175.

Mullainathan, S., & Shafir, E. (2013). Decision making and policy in contexts of poverty. In Eldar Shafir (Ed.), *The behavioral foundations of public policy*. Princeton, NJ: Princeton University Press.

Mullainathan, S., & Shafir, E. (2014). *Scarcity: The new science of having less and how it defines our lives* (Reprint ed.). New York, NY: Picador.

Nair, S., & Howlett, M. (2015). **Scaling up of policy experiments and pilots: A qualitative comparative analysis and lessons for the water sector**. *Water Resources Management*, *29*(14), 4945–4961.

Nickerson, R. S. (1998). Confirmation bias: A ubiquitous phenomenon in many guises. *Review of General Psychology*, *2*(2), 175.

Nielsen, A. S. E., Sand, H., Sørensen, P., Knutsson, M., Martinsson, P., Persson, E., & Wollbrant, C. (2017). ***Nudging and pro-environmental behaviour***. Nordic Council of Ministers.

Nordhaus, W. D. (2015). *The climate casino: Risk, uncertainty, and economics for a warming world*. New Haven, CT: Yale University Press.

O'Donoghue, T., & Rabin, M. (1999). Doing it now or later. *American Economic Review*, *89*(1), 103–124.

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date: 30 July 2019

Office of Gas and Electricity Markets. (2011). ***Energy demand research project final analysis***.

Oliver, A. (2013). **From nudging to budging: Using behavioural economics to inform public sector policy**. *Journal of Social Policy*, *42*(4), 685–700.

Organisation for Economic Co-operation and Development. (2017a). ***Behavioural insights and public policy***. OECD Publishing.

Organisation for Economic Co-operation and Development. (2017b). ***Tackling environmental problems with the help of behavioural insights***. OECD Publishing.

Pacala, S., & Socolow, R. (2004). Stabilization wedges: Solving the climate problem for the next 50 years with current technologies. *Science*, *305*(5686), 968–972.

Pallak, M. S., & Cummings, W. (1976). **Commitment and voluntary energy conservation**. *Personality and Social Psychology Bulletin*, *2*(1), 27–30.

Patt, A. G., & Schröter, D. (2008). Perceptions of climate risk in Mozambique: Implications for the success of adaptation strategies. *Global Environmental Change*, *18*(3), 458–467.

Perez-Batres, L. A., Miller, V. V., & Pisani, M. J. (2011). **Institutionalizing sustainability: An empirical study of corporate registration and commitment to the United Nations global compact guidelines**. *Journal of Cleaner Production*, *19*(8), 843–851.

Pew Research Center. (2009). *Modest support for "cap and trade" policy: Fewer Americans see solid evidence of global warming*. Washington, DC: Pew Research Center.

Pichert, D., & Katsikopoulos, K. V. (2008). **Green defaults: Information presentation and pro-environmental behaviour**. *Journal of Environmental Psychology*, *28*(1), 63–73.

Pollitt, M. G., & Shaorshadze, I. (2011). ***The role of behavioural economics in energy and climate policy***. Cambridge, UK: University of Cambridge.

Reisch, L. A., & Sunstein, C. R. (2016). Do Europeans like nudges? *Judgement and Decision Making*, *11*(4), 310–325.

Rivas, C. S., Cuniberti, B., & Bertoldi, P. (2016). ***Information measures to promote energy use reduction across EU Member States***. EUR 27997 EN.

Schneider, A., & Ingram, H. (1990). **Behavioral assumptions of policy tools**. *Journal of Politics*, *52*(2), 510–529.

Schneider, A., & Ingram, H. (1993). **Social construction of target populations: Implications for politics and policy**. *American Political Science Review*, *87*(2), 334–347.

Schneider, A. L., & Ingram, H. (1997). *Policy design for democracy*. Lawrence: University Press of Kansas.

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date: 30 July 2019

Schroeder, H. (2010). The history of international climate change politics: Three decades of progress, process and procrastination. In M. Boykoff (Ed.), *Politics of climate change: A survey* (pp. 26–41). London, UK: Routledge.

Sethi-Iyengar, S., Huberman, G., & Jiang, G. (2004). **How much choice is too much? Contributions to 401(k) Retirement Plans**. In O. S. Mitchell & S. P. Utkus (Eds.), *Pension design and structure: New lessons from behavioural finance* (pp. 83–96). Oxford, UK: Oxford University Press.

Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and quasi-experimental designs for generalized causal inference*. Boston, MA: Houghton Mifflin.

Shah, A. K., & Oppenheimer, D. M. (2008). **Heuristics made easy: An effort-reduction framework**. *Psychological Bulletin*, *134*(2), 207–222.

Smit, B., & Wandel, J. (2006). **Adaptation, adaptive capacity and vulnerability**. *Global Environmental Change*, *16*(3), 282–292.

Smith, G., & Visser, M. (2014). *Behavioural nudges as a water savings strategy: Report to the Water Research Commission*. South Africa: Water Research Commission.

Smith, N., & Leiserowitz, A. (2014). **The role of emotion in global warming policy support and opposition**. *Risk Analysis*, *34*(5), 937–948.

Sousa Lourenço, J., Ciriolo, E., Rafael Almeida, S., & Troussard, X. (2016). ***Behavioural insights applied to policy: European report 2016***. Luxembourg, Belgium: European Commission Joint Research Centre.

Spence, A., Poortinga, W., Butler, C., & Pidgeon, N. F. (2011). **Perceptions of climate change and willingness to save energy related to flood experience**. *Nature Climate Change*, *1*(1), 46–49.

Stadelmann, M., & Schubert, R. (2018). **How do different designs of energy labels influence purchases of household appliances? A field study in Switzerland**. *Ecological Economics*, *144*, 112–123.

Staub, E. (1972). **Instigation to goodness: The role of social norms and interpersonal influence**. *Journal of Social Issues*, *28*(3), 131–150.

Stern, N. H. (2007). *The economics of climate change: The Stern review*. Cambridge, UK: Cambridge University Press.

Stewart, J. (1993). **Rational choice theory, public policy and the liberal state**. *Policy Sciences*, *26*(4), 317–330.

Stone, D. (2001). *Policy paradox: The art of political decision making* (3rd ed.). New York, NY: W. W. Norton.

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date: 30 July 2019

Strassheim, H., & Korinek, R.-L. (2015). Behavioural governance in Europe. In J. Wilsdon & R. Doubleday (Eds.), *Future directions for scientific advice in Europe* (pp. 155–162). Cambridge, UK: Centre for Science and Policy.

Strassheim, H., & Korinek, R.-L. (2016). **Cultivating "nudge": Behavioural governance in the UK**. In J.-P. Voß & R. Freeman (Eds.), *Knowing governance* (pp. 107–126). London, UK: Palgrave Macmillan UK.

Sunstein, C. R. (1996). **Social norms and social roles**. *Columbia Law Review*, *96*(4), 903–968.

Sunstein, C. R. (2006). **The availability heuristic, intuitive cost-benefit analysis, and climate change**. *Climatic Change*, *77*(1–2), 195–210.

Sunstein, C. R. (2013). *Simpler: The future of government* (Reprint ed.). New York, NY: Simon & Schuster.

Sunstein, C. R. (2014). **Nudging: A very short guide**. *Journal of Consumer Policy*, *37*(4), 583–588.

Sunstein, C. R. (2015). *Why nudge? The politics of libertarian paternalism* (Reprint ed.). New Haven, CT: Yale University Press.

Sunstein, C. R. (2016). *The ethics of influence: Government in the age of behavioral science*. New York, NY: Cambridge University Press.

Sunstein, C. R., & Reisch, L. A. (2013). **Green by default**. *Kyklos*, *66*(3), 398–402.

Sunstein, C. R., & Reisch, L. A. (2016). ***Climate-friendly default rules***. Discussion Paper No. No. 878. Cambridge, MA: Harvard Law School.

Sunstein, C. R., & Thaler, R. H. (2003). Libertarian paternalism is not an oxymoron. *University of Chicago Law Review*, *70*(4), 1159–1202.

Taniguchi, A., & Fujii, S. (2007). **Promoting public transport using marketing techniques in mobility management and verifying their quantitative effects**. *Transportation*, *34*(1), 37–49.

Taylor, M. (2008). **Beyond technology-push and demand-pull: Lessons from California's solar policy**. *Energy Economics*, *30*(6), 2829–2854.

Thaler, R. H. (2015). *Misbehaving: The making of behavioral economics*. New York, NY: W. W. Norton.

Thaler, R., & Mullainathan, S. (2008). **Behavioral economics**. In *The concise encyclopedia of economics* (2nd ed.).

Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving decisions about health, wealth, and happiness* (Revised & expanded ed.). New York, NY: Penguin Books.

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date: 30 July 2019

Tobin, P., Schmidt, N. M., Tosun, J., & Burns, C. (2018). **Mapping states' Paris climate pledges: Analysing targets and groups at COP 21**. *Global Environmental Change*, *48*, 11–21.

Torgerson, D. (1985). Contextual orientation in policy analysis: The contribution of Harold D. Lasswell. *Policy Sciences*, *18*(3), 241–261.

Tversky, A., & Kahneman, D. (1973). Availability: A heuristic for judging frequency and probability. *Cognitive Psychology*, *5*(2), 207–232.

Tversky, A., & Kahneman, D. (1974). **Judgment under uncertainty: Heuristics and biases**. *Science*, *185*(4157), 1124–1131.

Tversky, A., & Kahneman, D. (1981). **The framing of decisions and the psychology of choice**. *Science*, *211*(4481), 453–458.

Ungar, S. (1992). The rise and (relative) decline of global warming as a social problem. *Sociological Quarterly*, *33*(4), 483–501.

United Nations. (1992). **United Nations framework convention on climate change**.

Valatin, G., Moseley, D., & Dandy, N. (2016). **Insights from behavioural economics for forest economics and environmental policy: Potential nudges to encourage woodland creation for climate change mitigation and adaptation?** *Forest Policy and Economics*, *72*, 27–36.

van der Linden, S. (2015). **Exploring beliefs about bottled water and intentions to reduce consumption: The dual-effect of social norm activation and persuasive information**. *Environment and Behavior*, *47*(5), 526–550.

Viscusi, W. K., & Gayer, T. (2015). Behavioral public choice: The behavioral paradox of government policy. *Harvard Journal of Law & Public Policy*, *38*, 973.

Vliet, B., van, Chappells, H., & Shove, E. (2005). *Infrastructures of consumption: Environmental innovation in the utility industries*. London, UK: Earthscan.

Vreugdenhil, H., Taljaard, S., & Slinger, J. H. (2012). **Pilot projects and their diffusion: A case study of integrated coastal management in South Africa**. *International Journal of Sustainable Development*, *15*(1–2), 148.

Weaver, R. K. (2009). ***Target compliance: The final frontier of policy implementation***. Issues in Governance Studies No. 27. Brookings Institution.

Weaver, R. K. (2014). **Compliance regimes and barriers to behavioral change**. *Governance*, *27*(2), 243–265.

Weaver, R. K. (2015). **Getting people to behave: Research lessons for policy makers**. *Public Administration Review*, *75*(6), 806–816.

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date: 30 July 2019

Weber, E. U. (2016). **What shapes perceptions of climate change? New research since 2010: What shapes perceptions of climate change?** *Wiley Interdisciplinary Reviews: Climate Change*, *7*(1), 125–134.

Weber, E. U., & Stern, P. C. (2011). Public understanding of climate change in the United States. *American Psychologist*, *66*(4), 315.

Werner, C. M., Turner, J., Shipman, K., Twitchell, F. S., Dickson, B. R., Bruschke, G. V., & von Bismarck, W. B. (1995). **Commitment, behavior, and attitude change: An analysis of voluntary recycling**. *Journal of Environmental Psychology*, *15*(3), 197–208.

White, M. D. (2016). **Overview of behavioral economics and policy**. In S. Abdukadirov (Ed.), *Nudge theory in action* (pp. 15–36). Cham, Switzerland: Springer International.

Whyte, G., & Levi, A. S. (1994). **The origins and function of the reference point in risky group decision making the case of the Cuban Missile Crisis**. *Journal of Behavioral Decision Making*, *7*(4), 243–260.

Wilbanks, T. J., Kane, S. M., Leiby, P. N., Perlack, R. D., Settle, C., Shogren, J. F., & Smith, J. B. (2003). **Possible responses to global climate change: Integrating mitigation and adaptation**. *Environment: Science and Policy for Sustainable Development*, *45*(5), 28–38.

Wilkinson, T. M. (2013). **Nudging and manipulation**. *Political Studies*, *61*(2), 341–355.

World Bank. (2010). ***World development report 2010: Development and climate change***. Washington, DC: World Bank.

World Bank. (2015). ***World development report 2015: Mind, society and behaviour***. Washington, DC: World Bank.

Zaval, L., & Cornwell, J. F. M. (2016). **Cognitive biases, non-rational judgments, and public perceptions of climate change**. *Oxford Research Encyclopedia*.

Zaval, L., Keenan, E. A., Johnson, E. J., & Weber, E. U. (2014). **How warm days increase belief in global warming**. *Nature Climate Change*, *4*(2), 143–147.

Notes:

(1.) According to IPCC (2014) climate mitigation policies aim to reduce or eliminate the drivers of climate change, while climate adaptation policies aim to manage or limit the actual or expected climate change impacts.

(2.) *Nudge* by Thaler and Sunstein (2008) is a popular book that focuses on the application of behavioral insights to policymaking. The authors define a nudge as “any aspect of the choice architecture that alters people’s behavior in a predictable way without forbidding any options or significantly altering their economic incentives” (p. 6). They use the term “choice architecture” to explain that by tapping into individuals’ cognitive biases,

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policy can be designed in a way to influence people's choices in a certain direction. They term this approach "libertarian paternalism" (Thaler & Sunstein, 2003, 2008).

(3.) IPCC and Stern speak of behavior in terms of individual choice or consumption and not specifically in terms of insights drawn from the behavioral sciences.

(4.) The two other essential elements for climate change policy identified by Stern (2007) are carbon pricing and technology policy.

(5.) Monitoring issues imply difficulty in monitoring compliance because of high costs or due to the nature of the activity, which may be widely dispersed or may take place in private. Enforcement issues may stem from weak enforcement by frontline staff. Incentive issues may be rooted in insufficient incentives for compliance due to hidden opportunity costs, inertia, and satisficing behavior of policy targets. Resource barriers may originate in a lack of resources to comply with the policy. Autonomy barriers may be due to a lack of autonomy of the policy-taker to be able to comply with the policy. Resource and autonomy barriers tend to be associated with lack of capacity to comply rather than a lack of willingness (Weaver, 2009, 2014, 2015).

(6.) Overconfidence effect refers to the observation that people's subjective confidence in their abilities is greater than their objective (actual) performance (Moore & Healy, 2008). Present bias refers to the preference that people exhibit for the present over the future (O'Donoghue & Rabin, 1999).

(7.) While such research centers are referred to as "nudge units" in common parlance, it may be remembered that a nudge is just one type of application of behavioral sciences. Thus these research centers adopt a broader view toward using findings from behavioral sciences in a wide variety and manner of policy instruments or tools, with nudges constituting only a part of this larger menu (Hallsworth, 2016).

(8.) Sunstein (2014) catalogs ten nudges considered important for public policy: (i) default rules; (ii) simplification; (iii) use of social norms; (iv) increases in ease and convenience; (v) disclosure; (vi) warnings, graphic or otherwise; (vii) pre-commitment strategies; (viii) reminders; (ix) eliciting implementation intentions; and (x) informing people of the nature and consequences of their own past choices. Another prominent intervention not included in Sunstein's list is that of making changes to the physical environment or interventions that alter the micro-environment with the aim of changing behavior. These have frequently been used in the health sector (see for instance Allan, Querstret, Banas, & de Bruin, 2017; Hollands et al., 2013).

(9.) Cognitive dissonance (Festinger, 1962) refers to the uncomfortable tension that occurs when an individual realizes that she or he has engaged in a behavior that is inconsistent with the type of person she or he would to be to be seen publicly to be—resulting in two conflicting and simultaneous and feelings or ideas.

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(10.) An individual’s economic behavior in relation to environmental policy is often seen in a dual frame—as a consumer (individualistic, responding to economic incentives) or as a citizen (team player, motivated by an altruistic concern for a larger community). Thus, construction of effective environmental policy requires focusing on the appropriate motivation that is more salient in the individual (Berglund & Matti, 2006).

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